

Effects of acute consumption of a fruit and vegetable purée-based drink on vasodilation and oxidative status.

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Epidemiological studies indicate that diets rich in fruits and vegetables (F&V) are protective against CVD. Puréed F&V products retain many beneficial components, including flavonoids, carotenoids, vitamin C and dietary fibres. The present study aimed to establish the physiological effects of acute ingestion of a F&V purée-based drink (FVPD) on vasodilation, antioxidant status, phytochemical bioavailability and other CVD risk factors. A total of twenty-four subjects, aged 30-70 years, completed the randomised, single-blind, controlled, crossover test meal study. Subjects consumed 400 ml of the FVPD, or a fruit-flavoured sugar-matched control, after following a low-flavonoid diet for 5 d. Blood and urine samples were collected throughout the study day, and vascular reactivity was assessed at 90 min intervals using laser Doppler iontophoresis. The FVPD significantly increased plasma vitamin C ($P=0.002$) and total nitrate/nitrite ($P=0.001$) concentrations. There was a near significant time by treatment effect on ex vivo LDL oxidation ($P=0.068$), with a longer lag phase after consuming the FVPD. During the 6 h after juice consumption, the antioxidant capacity of plasma increased significantly ($P=0.003$) and there was a simultaneous increase in plasma and urinary phenolic metabolites ($P<0.05$). There were significantly lower glucose and insulin peaks after ingestion of the FVPD compared with control ($P=0.019$ and 0.003) and a trend towards increased endothelium-dependent vasodilation following FVPD consumption ($P=0.061$). Overall, FVPD consumption significantly increased plasma vitamin C and total nitrate/nitrite concentrations, with a trend towards increased endothelium-dependent vasodilation. Puréed F&V products are useful vehicles for increasing micronutrient status, plasma antioxidant capacity and in vivo NO generation, which may contribute to CVD risk reduction.